

Programme Code: TU856-1, TU858-1

Module Code: CMPU 1031

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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

TU856 BSc. (Honours) Degree in Computer Science

TU858 BSc. (Honours) Degree in Computer Science
(International)
Year 1

SEMESTER 1
EXAMINATIONS 2023/24

CMPU 1031 Web Development 1

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Exam Duration: Two hours

Instructions:

Answer question (1) and any two of questions (2), (3), (4)

Question (1) is worth 40 marks.

Questions (2), (3), (4) are worth 30 marks each

1.(a) In Computer Science, Separation of Concerns is a design principle for separating a program into distinct sections. Explain how the separation of concerns is applied in Web Development. List and explain 3 (three) benefits of separating the HTML from the CSS code.

(10 marks)

1.(b) Explain what is a meta tag in HTML and how the search engine can use it to learn about the website content.

(5 marks)

1.(c) Briefly explain the benefits of using HTML5 semantic tags and how this can benefit users with accessibility issues.

(5 marks)

1.(d) What is the Document Object Model (DOM)? Give an example of how JavaScript can be used to manipulate the DOM.

(8 marks)

1.(e) The CSS box model represents how HTML elements are rendered in the browser. List and explain each of the four components of the box model. Provide a diagram to illustrate your answer.

(12 marks)

2.(a) Consider a HTML button element that has a class named “my-button”. Write a CSS code to:

- make the button change its background colour to blue when hovered over and back to its original colour when not hovered.
- make the text written in inside the button white.

(10 marks)

2.(b) Write a JavaScript function named multiply that takes two numbers as arguments and returns their product.

(10 marks)

2.(c) Explain the main differences between static and absolute CSS positioning.

(10 marks)

3.(a) Before developing a website, it is important to work on prototypes that demonstrate how the website is expected to look. Define what is a low-fidelity prototype and a high-fidelity prototype of a website.

(10 marks)

3.(b) Explain what Bootstrap is and provide two key benefits of using Bootstrap in web development.

(10 marks)

3.(c) Use an example to explain the basic operation of the descendant selector in CSS.
(10 marks)

4.(a) Explain the main differences between an absolute and a relative path, and provide an example for each of these.
(10 marks)

4.(b) Explain CSS media queries and how they can be used to make a website responsive when accessed from different devices.
(10 marks)

4.(c) The image tag requires a value for the attribute source, which will provide the image location. Consider that you need to insert an image tag in a file called index.html to include an image called penguin.jpg. Provide the necessary line of HTML code to include the image when:

(a.1) The penguin.jpg file is located in the same folder as the index.html file.

(a.2) The index.html file is located in the main folder, and the penguin.jpg file is located in a subfolder called images.

(6 marks)

4.(d) Write the HTML code to create an ordered list with three list items containing the words "Apple," "Banana," and "Cherry."
(4 marks)